
FULL STACK WEB DEVELOPMENT – EXPERIMENT RECORD

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Subject: FS

Experiment Title: HTML5 Canvas Graphics and Styled Webpage Using HTML5 and CSS3

1. OBJECTIVE / AIM

To develop a webpage using HTML5 Canvas to draw basic graphics such as a rectangle, circle, and line, and to design an attractive webpage using HTML5 and CSS3 features such as gradients, navigation menus, cards, hover effects, shadows, and a footer.

2. SYSTEM REQUIREMENTS

Hardware Requirements:

- Computer or Laptop
- Minimum 4 GB RAM
- Keyboard and Mouse

Software Requirements:

- Operating System: Windows / Linux
 - Code Editor: Visual Studio Code / Notepad
 - Web Browser: Google Chrome / Microsoft Edge / Mozilla Firefox
 - Technologies Used: HTML5, CSS3 and JavaScript
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3. ARCHITECTURE / DESIGN

This experiment consists of two parts.

Part A – HTML5 Canvas Graphics

The HTML5 Canvas element is used to create graphics dynamically using JavaScript. A 2D drawing context is obtained from the canvas, which is then used to draw different shapes and text.

The following elements are created:

- Rectangle using the fillRect() method
- Circle using the arc() method
- Line using moveTo() and lineTo() methods
- Text using the fillText() method

Part B – Styled Webpage

HTML5 is used to create the structure of the webpage, while CSS3 is used to improve its appearance and layout.

The webpage contains:

- Header with gradient background and text shadow
- Navigation bar with Home, About and Contact links
- Three content cards for HTML5, CSS3 and Web Design
- Learn More buttons with hover effects
- Footer containing copyright information

CSS properties such as linear-gradient(), border-radius, box-shadow, display: flex and :hover are used to make the webpage attractive.

4. SOURCE CODE

PART A – HTML5 CANVAS GRAPHICS

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>HTML5 Canvas</title>
```

```
</head>
```

```
<body>
```

```
<h1>HTML5 Canvas Graphics</h1>
```

```
<canvas id="canvas" width="500" height="250"
```

```
style="border:1px solid black;"></canvas>
```

```
<script>
```

```
let c = document.getElementById("canvas");
```

```
let x = c.getContext("2d");
```

```
// Rectangle
```

```
x.fillStyle = "royalblue";
```

```
x.fillRect(30, 30, 100, 60);
```

```
// Circle
```

```
x.beginPath();
```

```
x.arc(220, 60, 30, 0, 2 * Math.PI);
```

```
x.stroke();
```

```
// Line
```

```
x.beginPath();
```

```
x.moveTo(300, 30);
```

```
x.lineTo(420, 100);
```

```
x.stroke();
```

```
// Name
```

```
x.font = "22px Arial";
```

```
x.fillText("Isha", 200, 180);
```

```
</script>
```

```
</body>
```

```
</html>
```

PART B – MY STYLED WEBPAGE

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>My Styled Webpage</title>
```

```
  <style>
```

```
    body {
```

```
      margin: 0;
```

```
      font-family: Arial;
```

```
      background: #eee;
```

```
    }
```

```
    /* Header */
```

```
    header {
```

```
      padding: 20px;
```

```
      text-align: center;
```

```
      color: white;
```

```
      background: linear-gradient(to right, blue, skyblue);
```

```
      text-shadow: 2px 2px 3px black;
```

```
    }
```

`/* Navigation */`

```
nav {  
  
    padding: 15px;  
  
    text-align: center;  
  
    background: #222;  
  
}
```

```
nav a {  
  
    color: white;  
  
    margin: 15px;  
  
    text-decoration: none;  
  
}
```

```
nav a:hover {  
  
    color: yellow;  
  
}
```

`/* Cards */`

```
.cards {  
  
    display: flex;  
  
    gap: 20px;  
  
    margin: 30px;  
  
}
```

```
.card {  
    padding: 20px;  
    background: white;  
    border: 2px solid blue;  
    border-radius: 12px;  
    box-shadow: 3px 3px 8px gray;  
    text-align: center;  
}
```

```
button {  
    padding: 8px 15px;  
    border: 0;  
    border-radius: 6px;  
    background: blue;  
    color: white;  
}
```

```
button:hover {  
    background: black;  
}
```

```
/* Footer */
```

```
footer {  
    padding: 15px;  
    text-align: center;
```

background: #222;

color: white;

}

</style>

</head>

<body>

<header>

<h1>My Styled Webpage</h1>

</header>

<nav>

Home

About

Contact

</nav>

<div class="cards">

<div class="card">

<h2>HTML5</h2>

<p>Creates webpage structure.</p>

<button>Learn More</button>

</div>

```
<div class="card">
```

```
  <h2>CSS3</h2>
```

```
  <p>Styles the webpage.</p>
```

```
  <button>Learn More</button>
```

```
</div>
```

```
<div class="card">
```

```
  <h2>Web Design</h2>
```

```
  <p>Makes websites attractive.</p>
```

```
  <button>Learn More</button>
```

```
</div>
```

```
</div>
```

```
<footer>
```

```
  <p>© 2026 My Styled Webpage</p>
```

```
</footer>
```

```
</body>
```

```
</html>
```

5. COMPILATION / EXECUTION STEPS

1. Open Visual Studio Code or any suitable text editor.
2. Create a new HTML file named **partA.html**.
3. Enter the Part A HTML and JavaScript code and save the file.

4. Open the **partA.html** file in a web browser.
5. Observe the rectangle, circle, line and text displayed inside the canvas.
6. Create another HTML file named **partB.html**.
7. Enter the Part B HTML and CSS code and save the file.
8. Open the **partB.html** file in a web browser.
9. Observe the header, navigation bar, content cards, buttons and footer.
10. Move the mouse over the navigation links and buttons to observe the hover effects.

Note: No separate compilation is required. HTML, CSS and JavaScript can be executed directly using a modern web browser.

6. EXPECTED OUTPUT / SCREENSHOT

PART A – HTML5 CANVAS GRAPHICS

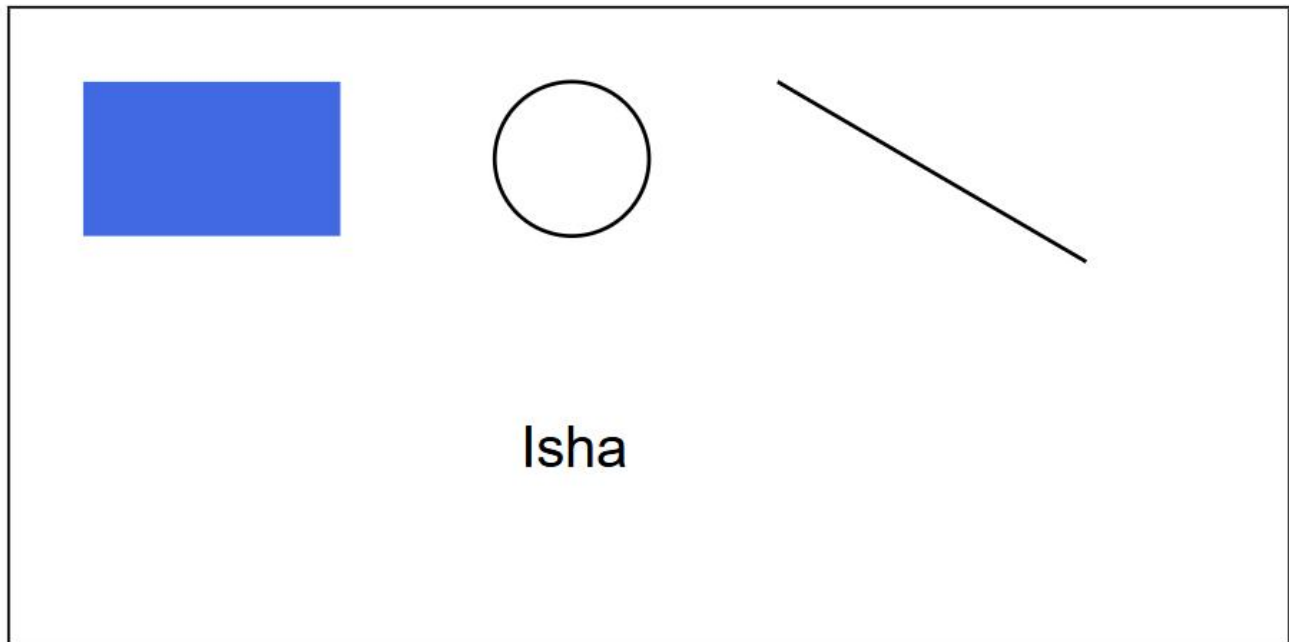
The output displays a heading named **HTML5 Canvas Graphics** followed by a canvas area with a black border.

The canvas contains:

- A royal-blue rectangle
- An outlined circle
- A diagonal line
- The text “Isha”

Screenshot – Part A:

HTML5 Canvas Graphics

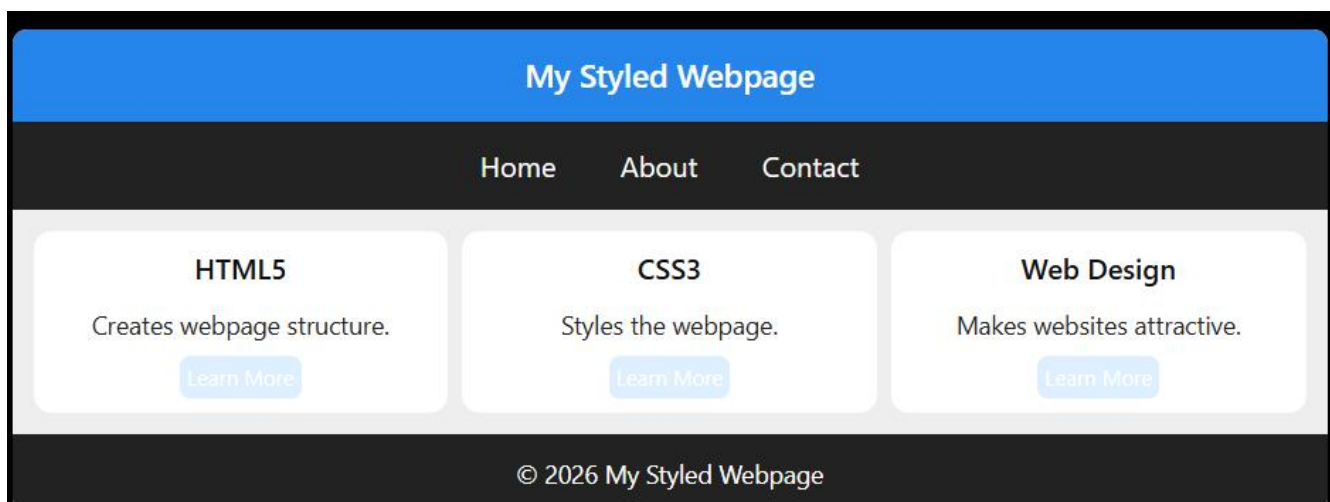


PART B – MY STYLED WEBPAGE

The output displays:

- A blue gradient header with the title “My Styled Webpage”
- A dark navigation bar containing Home, About and Contact links
- Three styled cards for HTML5, CSS3 and Web Design
- Learn More buttons
- A dark footer with copyright information

Screenshot – Part B:



7. RESULT / CONCLUSION

Thus, the HTML5 Canvas graphics were successfully created using JavaScript to draw a rectangle, circle, line and text. A styled webpage was also successfully designed using HTML5 and CSS3 with a gradient header, navigation bar, content cards, hover effects, buttons, shadows and footer. The experiment demonstrates the practical use of HTML5, CSS3 and JavaScript in developing interactive and attractive web pages.
